

THE DUKE'S COURTROOM

MAJOR EVIDENCE CHUTE

MINOR EVIDENCE CHUTE

MICRO

×3 SETS
×3 10 mL
TYPICAL ≥2 SETS

Q IgG ≥1:800
BART IgG ≥1:800

IMAGING

ECHO VEGETATION CT SCAN, WARNING: PET ABSCESS FALSE+ <3 MO

SURGICAL

MINOR EVIDENCE CHUTE

>38°C

skin + nodule
retinal spot

PET culture <3 MO

murmur

DEFINITE

2 MAJOR
1 MAJ + 3 MIN
5 MINOR

POSSIBLE

1 MAJ + 1 MIN
3 MINOR

→ TREAT

ECHO STATION

MISSES 20-35%

>90%

NEG → REPEAT 7-10 DAYS
TEE IF ANY RISK

TREATMENT APOTHECARY

SHELF 1

4 WKS

short-course: +
2 WKS EACH

NOT PVE

CLEAR: 2 DAYS

SHELF 2

GOLD FRAME

AMP ×6 WKS

LESS NEPHROTOXIC
COVERS HILGR
E. FAECALIS

NOT E. FAECIUM

pencil + arrow backup
4-6 WKS

BOARD TRAP

NOT BACTERICIDAL

CLEAR: 2 DAYS

SHELF 3

MSSA NVE

NAFCILLIN 4-6 WKS

PROMINENT

LARGEST

2 WKS

CLEAR: 3-5 DAYS

MRSA NVE

VAN + dart 18 mg/kg 4-6 WKS

PVE

BOARD TRAP

TRAP DOOR HVISA: MIC 2

HIDDEN LURKING BENEATH

NOT DETECTABLE

VISA: MIC 4-8

VISUALLY LOUD

Nafcillin/VAN + arrow + RIFAMPIN BULLET BIOFILM

6-8 WKS

LF 4

4 WKS (6 PVE)

18 MO / 24 MO

6 WKS + 2 WKS

mushroom: amphibian ± pill

EARLY SURGERY

INDEFINITE

LF 5

'ACUTE': VAN + arrow/cefepime

SUBACUTE': VAN + axe

PVE ≤1 YR': VAN + arrow + cefepime

MONITORING STATION

DAILY CULTURES → STERILE
FEVER ≥7 DAYS → EVALUATE
DURATION FROM NEGATIVE PVE = LONGER

1. Duke major — blood cultures

WHAT YOU SEE

MICRO chute, x2/x3 vials, typical ≥ 2 sets

WHAT IT ENCODES

Major criterion: typical organism in 2 separate cultures, or persistently positive cultures. Draw at least 3 sets from different sites before antibiotics.

BOARD TRAP

A single positive culture (esp. coag-negative staph) may be contaminant — persistence is what makes it a major criterion.

2. Coxiella serology — major

WHAT YOU SEE

Q IgG titer $\geq 1:800$

WHAT IT ENCODES

A single Coxiella burnetii (Q fever) antibody titer $\geq 1:800$ or positive PCR counts as a major microbiologic criterion in culture-negative IE.

BOARD TRAP

Culture-negative IE still meets a major criterion via Q-fever serology — don't abandon the diagnosis on negative cultures.

3. Imaging — echo vegetation

WHAT YOU SEE

IMAGING — echo vegetation/abscess, CT

WHAT IT ENCODES

Major imaging criterion: vegetation, abscess, or new valve dehiscence on echo. TTE first; TEE is far more sensitive for vegetations, abscess, and prosthetic valves.

BOARD TRAP

A negative TTE does not rule out IE — proceed to TEE when suspicion is high.

4. PET — prosthetic timing

WHAT YOU SEE

WARNING PET false if < 3 months

WHAT IT ENCODES

FDG-PET/CT can support prosthetic valve IE, but is falsely positive in the first ~ 3 months after surgery due to postoperative inflammation.

BOARD TRAP

Positive PET within 3 months of valve surgery may be post-op inflammation, not infection.

5. Definite vs possible

WHAT YOU SEE

Scale — DEFINITE 2 major / 1+3 / 5 minor; POSSIBLE

WHAT IT ENCODES

Definite IE: 2 major, or 1 major + 3 minor, or 5 minor. Possible IE: 1 major + 1 minor, or 3 minor. Minor criteria include fever, predisposition, vascular and immunologic phenomena.

BOARD TRAP

Pathologic confirmation (positive vegetation culture/histology) is automatically definite regardless of clinical tally.

6. Empiric / suspicion to treat

WHAT YOU SEE

ECHO STATION — neg repeat 7-10 days, TEE if any risk

WHAT IT ENCODES

If initial echo is negative but suspicion persists, repeat in 7-10 days and use TEE. Sensitivity gaps mean missed cases on a single early study.

BOARD TRAP

Stopping the workup after one negative TTE misses 20-35% of vegetations.

7. Native MSSA — nafcillin

WHAT YOU SEE

Shelf — MSSA NVE, nafcillin 4-6 wks

WHAT IT ENCODES

Methicillin-susceptible S. aureus native-valve IE: nafcillin or cefazolin for 4-6 weeks. Beta-lactam outperforms vancomycin for MSSA.

BOARD TRAP

Using vancomycin for proven MSSA is inferior — switch to nafcillin/cefazolin once susceptibility is known.

8. Native MRSA — vancomycin

WHAT YOU SEE

MRSA NVE, vanc, MIC ≤ 1

WHAT IT ENCODES

MRSA native-valve IE: vancomycin (or daptomycin) for ~ 6 weeks; target appropriate vancomycin levels (MIC ≤ 1 favorable).

BOARD TRAP

Daptomycin is an alternative but is inactivated by pulmonary surfactant — do not use it for pneumonia.

9. Viridans strep — penicillin

WHAT YOU SEE

LF5 SUBACUTE: VAN + axc; penicillin/ceftriaxone

WHAT IT ENCODES

Penicillin-susceptible viridans/strep IE: penicillin or ceftriaxone 4 weeks (2 weeks with gentamicin synergy in selected cases).

BOARD TRAP

Ceftriaxone once-daily enables outpatient therapy for susceptible strep — overtreating with broad agents is unnecessary.

10. Enterococcus — synergy

WHAT YOU SEE

LF5 ACUTE/empiric: VAN + arrow/cefepime

WHAT IT ENCODES

Enterococcal IE needs cell-wall agent plus synergy: ampicillin + ceftriaxone (or aminoglycoside). Long course given relative resistance.

BOARD TRAP

Monotherapy fails enterococcal IE — synergistic dual therapy is required.

11. Prosthetic valve + rifampin

WHAT YOU SEE

Nafcillin/VAN + RIFAMPIN bullet, biofilm

WHAT IT ENCODES

Prosthetic valve staph IE: add rifampin (biofilm penetration) plus gentamicin to vancomycin/nafcillin, for 6+ weeks. PVE <1 yr suggests nosocomial organisms.

BOARD TRAP

Rifampin alone breeds resistance — only add it as part of combination after bacteremia clears.

12. Surgery indications

WHAT YOU SEE

Shelf — heart failure, abscess, amphibian + pill, early surgery

WHAT IT ENCODES

Surgery for heart failure from valve destruction, perivalvular abscess, large mobile vegetations with emboli, fungal/resistant organisms, and persistent bacteremia despite therapy.

BOARD TRAP

Delaying surgery in IE-related heart failure or root abscess increases mortality — operate early.

13. Treatment duration counter

WHAT YOU SEE

LF4 — 4 wks/6 wks, 18mo/24mo cultures

WHAT IT ENCODES

Most native-valve IE is 4-6 weeks IV; prosthetic and fastidious (*Coxiella*) infections need months. Document negative cultures and timed clearance.

BOARD TRAP

Short courses cause relapse — duration is organism- and valve-specific, not one-size-fits-all.

14. Subacute vs acute approach

WHAT YOU SEE

LF5 banners ACUTE / SUBACUTE / PVE

WHAT IT ENCODES

Acute IE (*S. aureus*): cover staph empirically with vancomycin + cefepime. Subacute: can sometimes await cultures if stable. PVE <1 yr: vanc + gentamicin + rifampin + cefepime.

BOARD TRAP

Withholding empiric coverage in toxic acute IE awaiting cultures is dangerous — treat after drawing sets.
